

Sovereign Trace Protocol:

A Cryptographic Infrastructure for Permanent Significance

Registration Across Plural Civilizational Time Systems

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Abstract

This paper introduces the **Sovereign Trace Protocol (STP)** — a cryptographic infrastructure for permanent, computationally tamper-evident registration of significant moments across three civilizational time systems simultaneously: Gregorian, Hebrew lunisolar, and 13 Moon Dreamspell. STP operates at two scales with categorically different epistemic foundations. At the individual scale, it provides a mechanism for resolving what we term *significance hunger* — a proposed construct for the need to permanently register one's exact observations in time — without requiring an audience, platform, or institutional permission. This psychological claim is currently unverified and is explicitly marked as an open empirical question. At the organizational scale, it provides computationally tamper-evident audit infrastructure for AI systems: sealed failure logs, append-only remediation chains, and a formally defined **Epistemic Debt Score (EDS)** operationalizing AI epistemic integrity as: documenting failures honestly, resolving them at a measured rate, and improving over time. The core cryptographic mechanism is a SHA-256 seal binding entry content to all three calendar representations simultaneously, with optional Ed25519 signature and optional multi-chain blockchain anchoring. We describe the FROZEN-4.0 implementation (82 self-test checks passing, zero external dependencies), a 22-template submission layer, a six-tier certification system, and the full EDS formula. We identify four open empirical questions, declare all known architectural vulnerabilities and platform dependencies honestly, and map STP to related work in personal digital timestamping, legal standing of cryptographic seals, and AI audit infrastructure. The paper advances two structural claims: (1) [?] the hunger to register significance does not require an audience — it requires permanent, verifiable registration (unverified, empirically falsifiable); (2) [S] an organization with a documented, growing, improving failure history is more auditably trustworthy than one with no failure record.

Keywords: *cryptographic timestamping, epistemic integrity, AI audit infrastructure, calendar pluralism, significance registration, permanence infrastructure, SHA-256, epistemic debt, personal lifelogging*

1. Introduction

The problem of permanence for human significance has historically required institutional intermediaries: publishers, notaries, registrars, archives. A researcher must submit to peer review. An artist must be exhibited. A whistleblower must find a journalist. The *significance* of a moment — what one observed, decided, predicted, or experienced — has no native permanence mechanism for individuals acting outside institutions.

The parallel problem in AI governance is structural: organizations deploying AI systems have no external pressure to maintain honest records of failures. A failure resolved silently and never documented leaves no trace. A vendor who patches a critical AI output error without logging it has, from the record's perspective, a clean history. The absence of a failure ledger is indistinguishable from the absence of failures.

The Sovereign Trace Protocol addresses both problems with the same cryptographic mechanism. This is not accidental. The containment surface — what must stay inside the protocol's scope and what must stay outside — is structurally identical at both scales: **inside** is permanent registration of what is true; **outside** is any mechanism that requires external validation, social metrics, or editable records.

1.1 Epistemic Asymmetry Between Use Cases

[R] The two use cases share a cryptographic mechanism but rest on categorically different epistemic foundations, and this asymmetry must be declared at the outset rather than resolved by labeling them "dual-use."

The **individual use case** rests on a psychological claim — that the act of permanent registration resolves a particular kind of recognition need without requiring an audience — that is currently unverified. [?] It is marked throughout this paper with [?] and is the subject of the first open empirical question in Section 11.

The **organizational use case** rests on a structural argument about accountability and trust — that documented, improving failure records are more auditable than silence — that is [S]-tagged as a directional claim supported by the logic of epistemic transparency but not yet empirically validated at scale.

Both claims are presented without inflating their certainty. The cryptographic mechanism ([R]) is the strongest claim in this paper.

1.2 Motivation

The individual motivation for STP emerged from an observation about recognition: the hunger to register one's significance may resolve not when someone reads the record, but when the record becomes permanently true. A sealed cryptographic stamp that binds a moment to three time systems is not a means to an audience. It is proposed as the end. Whether this holds empirically is Q1 in Section 11.

The organizational motivation is structurally independent. An AI failure sealed before remediation begins is a permanent honest record of the organization's actual epistemic state at that moment. It cannot be edited. The SHA-256 seal breaks if the content changes. This is not surveillance — it is self-certification infrastructure for organizations that want to demonstrate epistemic integrity over time, not just claim it. [S]

1.3 Scope and Contributions

This paper makes the following contributions:

- A formal specification of the STP triple-time seal mechanism and its cryptographic properties, including all known vulnerabilities and their current mitigation status.
- A fully specified Epistemic Debt Score (EDS) — a 0–100 metric with declared formulas for all five components, falsification conditions, and adversarial attack surface.
- A description of the FROZEN-4.0 implementation: design constraints, API surface, retirement protocol, and archival policy.
- A 22-template submission layer covering individual, organizational, and institutional use cases, with declared platform dependency.
- A six-tier certification system with intake rules, auditor vetting criteria, and a disputes protocol.
- Discussion of prior art in personal digital timestamping, AI audit infrastructure, and legal standing of cryptographic seals.
- Four open empirical questions and a prioritized future build sequence.

This v1.2 revision was produced through a structured red-team process (PDE v0.5 + MPTE v1.1) covering 48 distinct findings across gaps, vulnerabilities, risks, weaknesses, and definitional inconsistencies. Every finding and its resolution is logged in **Appendix E**. Findings are referenced inline by ID (e.g., F-007) at the point in the text where the resolution applies.

2. Background and Related Work

2.1 Cryptographic Timestamping

Cryptographic timestamping has a well-established literature. Haber and Stornetta (1991) introduced the foundational idea of linking document hashes in a chain to prevent backdating — work that directly preceded blockchain architecture. Bitcoin's proof-of-work mechanism (Nakamoto, 2008) extended this to a trustless, distributed setting.

STP requires **zero trusted third parties** for seal generation. [D] The SHA-256 seal is computed locally, offline, from Python's standard library, and is verifiable by any party with Python 3.11+ and the public parameters of the seal. Third-party blockchain anchoring (Bitcoin, Hedera, Ethereum) is available as a planned optional overlay — not a dependency of the primary seal, and not yet deployed (Stage 3, Section 12).

2.2 Calendar Systems as Epistemic Infrastructure

The choice to bind three calendar systems simultaneously is not arbitrary. Each encodes a different claim about the nature of time. However, the three systems are not equivalent in cultural adoption or institutional standing, and this asymmetry must be declared. [D]

The **Gregorian calendar** encodes civic time — the temporal infrastructure of current legal, financial, and institutional systems. A Gregorian timestamp is legible to courts, regulators, and employers. It is the global civic standard. [D]

The **Hebrew lunisolar calendar** encodes theological-historical continuity. At 5,786 years of counted time (at the 2026 anchor), it is one of the longest continuously maintained human time-reckoning systems. It is used for religious and some civic purposes by Jewish communities worldwide. [D]

The **13 Moon Dreamspell calendar** encodes rhythmic time — 13 moons of 28 days each, plus one Day Out of Time. Created in 1990 by José Argüelles, it is not a standard civic or religious calendar and is not used for any legal or institutional purpose by any significant population. [D] Its inclusion in STP reflects the protocol's position that cultural sovereignty over one's temporal record benefits from pluralism of time systems — but the Dreamspell's limited adoption means the "culturally sovereign across cultures" claim [S] carries significantly weaker empirical grounding than the Gregorian and Hebrew inclusions. Whether the Dreamspell's inclusion narrows the addressable population is an open empirical question (Q3, Section 11).

2.3 AI Audit Infrastructure

The AI audit infrastructure literature includes frameworks for internal, independent, and regulatory audits of AI systems, as well as regulatory requirements such as the EU AI Act (2024). STP's organizational use case is adjacent to but distinct from these frameworks. [?] A formal comparison of STP's EDS against established AI audit methodologies has not been performed and is declared as an open research question (Q4, Section 11).

STP does not assess AI system quality, fairness, or capability. It assesses **epistemic honesty**, operationally defined here as: the degree to which an organization maintains a tamper-evident, append-only record of AI system failures, resolves them at a documented rate, and demonstrates improvement over time. [S] This is not an established term of art — it is STP's working definition, declared for precision.

The Epistemic Debt Score is designed to be independent of AI system quality: an organization with a mature, honest failure ledger may score higher than one with a cleaner system and no ledger, because the absence of logged failures cannot be distinguished from the absence of failures.

2.4 Significance Hunger and Psychological Permanence

The term **significance hunger** as used in this paper refers to a specific proposed construct: the need to permanently register one's exact observations, decisions, or experiences in a form that is not contingent on another's reading or acknowledgment. This construct is original to STP and has not been validated against the psychological literature. [?]

Related constructs exist in existential psychology — meaning-making, symbolic immortality, the need for permanence — but no direct analogue to significance hunger as defined here (registration without audience as sufficient resolution) has been identified in the literature reviewed. Whether this construct maps onto, extends, or contradicts existing psychological frameworks is an open empirical question (Q1, Section 11).

STP does not claim that its individual use case is grounded in validated psychology. The specific claim — that permanent registration without audience resolves the significance need — requires dedicated empirical research before it can be treated as anything stronger than a directional hypothesis. [?]

2.5 Prior Art in Personal Digital Timestamping

Personal digital record-keeping and lifelogging systems have been developed by others prior to STP. [?] A systematic review of this prior art has not been performed by the author. STP's distinguishing architectural claims relative to any prior work are: the triple-time civilizational binding (anchoring simultaneously to three calendar systems), the dual-scale deployment (individual and organizational with a shared mechanism), and the Epistemic Debt Score as a formal organizational metric. [R] Whether these features are genuinely novel relative to all prior work is an open question the author cannot currently resolve.

2.6 Legal Standing of Cryptographic Seals

A significant limitation of STP's current specification is that the legal standing of its seals as evidence has not been analyzed, and this paper does not resolve that analysis. [?]

STP seals use no trusted third party and no certificate infrastructure. Whether an STP seal would be accepted as evidence in any jurisdiction — EU, US, or otherwise — depends on authentication procedures, chain of custody arguments, and specific legal context. This is unresolved and the author has not performed legal research on this question. [?]

The use case table in Section 8 references applications (whistleblower evidence, clinical records, scope disputes) where legal standing could be material. Users in these contexts should obtain independent legal advice and should not rely on STP seals as equivalent to notarized or legally qualified timestamps. [S]

3. Core Architectural Claims

STP rests on three foundational claims. Each is tagged per the ECF epistemic framework (Appendix B).

Claim 1 (C1): [R] The SHA-256 seal, when computed deterministically from entry text and triple-time parameters, produces a computationally tamper-evident binding of content to moment. Any modification to either the content or any timestamp field produces a different hash, detectable by recomputation. *Falsification test: demonstrate a collision — two distinct payloads producing the same SHA-256 output. No practical collision is known as of June 2026.*

Claim 2 (C2): [?] The permanent registration of a significant moment resolves the significance hunger at the moment of stamping — not at the moment of reading. This claim is the psychological core of STP's individual use case. It is currently unverified and requires empirical research to confirm. *Falsification test: a controlled study in which users who seal entries report no subjective difference in significance satisfaction compared to unsealed journaling. FCL-ST-001 (Appendix C) is the first empirical candidate.*

Claim 3 (C3): [S] An organization with a documented AI failure history — logged honestly, resolved at a measured rate, demonstrating improvement over time — is more auditably trustworthy than one with no failure record. *Falsification test: an organization with $EDS \geq 75$ experiences a catastrophic AI failure that its ledger shows no predecessor signals for — demonstrating that EDS scores do not predict epistemic readiness. Adversarial attack path: see Section 6.5.*

3.1 The TOPOS Shape Declaration

Every STP entry rests on a persistent architectural shape:

Containment surface: Inside — individual sovereignty over one's own temporal trace, immutable registration of significance across plural civilizational time systems, zero external validation required for resolution. Outside — social metrics, aggregation, ranking, editable records, platform dependency, any mechanism that requires another person's action to make the entry valid.

Note on platform dependency: STP's containment surface declares platform dependency as "outside." In the current FROZEN-4.0 implementation, the submission layer (Section 8) depends on GitHub, and the cryptographic linkage between failure and remediation records depends on ledger availability. These

dependencies are inside the current implementation but outside the containment surface's aspiration. The gap between declared containment and current implementation is acknowledged explicitly (see F-018, F-037 resolutions) and is the target of Stage 2 and Stage 3 (Section 12). [S]

Steering geometry: Every element directs toward permanence as the act of recognition itself. The stamp is not a means to an end. The stamp is the end. Registration equals resolution. Nothing downstream is required.

4. The Triple-Time Seal: Formal Specification

4.1 Payload Structure

The FROZEN-4.0 seal payload is a deterministically serialized JSON object with the following fields:

```
{
  "entry_text": "&lt;NFC-normalized UTF-8 string&gt;",
  "gregorian": "&lt;YYYY-MM-DD&gt;",
  "hebrew": "&lt;DD MonthName YYYY&gt;",
  "dreamspell": "Day &lt;N&gt;, &lt;Moon Name&gt; Moon &lt;M&gt;/13",
  "unix_utc": "&lt;integer seconds since epoch&gt;",
  "version": "FROZEN-4.0"
}
```

Serialization: `json.dumps(..., sort_keys=True, ensure_ascii=True)` — guarantees determinism across Python implementations and platforms. [D]

4.2 Seal Computation and Computational Permanence

```
seal = SHA-256(UTF-8-encode(json.dumps(payload, sort_keys=True, ensure_ascii=True)))
```

The seal is a 64-character hexadecimal string. It is **computationally bound** to all six fields simultaneously under SHA-256's collision resistance assumption. [R]

Computational vs. physical permanence: Throughout this paper, "permanent" means *computationally permanent under current cryptographic assumptions* — specifically, that SHA-256 is collision-resistant and preimage-resistant under classical computation. [D] SHA-256 is not proven post-quantum resistant. Grover's algorithm provides a quadratic speedup for preimage search on SHA-256, reducing effective security from 256 to approximately 128 bits under a quantum adversary. For an individual personal trace entry, 128-bit post-quantum security is adequate. For high-stakes organizational use cases over multi-decade timescales, post-quantum migration planning is advisable. [R]

Any modification to any payload field — including the version string — produces a different seal. [D]

4.3 Optional Signature Layer and Key Management

An Ed25519 signature may be appended to the seal record:

```
signature = Ed25519_sign(private_key, seal_bytes)
```

The signature provides authorship binding: proof that the sealing party held a specific private key at the time of stamping. The seal is valid without the signature; the signature adds non-repudiation.

Key management guidance [S]: The private key should be generated using a cryptographically secure random number generator, stored in an encrypted key store (e.g., hardware security module for organizational use, encrypted file vault for individual use), and never transmitted. The corresponding public key should be published in a verifiable location (e.g., committed to the public ledger at first use). Key rotation invalidates authorship binding for records signed under the previous key but does not invalidate the underlying SHA-256 seal. Lost private keys cannot be recovered; the seal remains valid but authorship cannot be proved. Key management at scale is a declared open architectural problem for STP Stage 2.

4.4 Calendar Implementations

Gregorian: Standard ISO 8601 date. Pre-reform boundary guard: dates before October 15, 1582 are rejected with an explicit error (the Gregorian calendar did not exist before this date; earlier dates are Julian, producing incorrect Julian Day Number values). [D]

Hebrew lunisolar: Implemented using the full four *dehiyot* (postponement rules) and civil-day convention. The four *dehiyot* are: (1) *Molad Zaken* — if the calculated new moon (*molad*) falls at or after noon; (2) *Lo ADU Rosh* — Rosh Hashanah may not fall on Sunday, Wednesday, or Friday; (3) *GaTaRaD* — specific condition on *molad* timing in common years; (4) *BeTU TeKaPoT* — specific condition on *molad* timing in years following leap years. The algorithm follows Dershowitz and Reingold (2018), whose *Calendrical Calculations* provides the definitive algorithmic treatment. All anchor dates verified against Hebcal.com, Chabad.org, and pyluach. [D]

13 Moon Dreamspell: The Dreamspell year begins on July 26 (Day 1, Moon 1, Magnetic Moon). The year consists of 13 moons of 28 days each (364 days), plus one Day Out of Time on July 25. Day number within moon and moon number are computed from the offset of the target date from the most recent July 26. [D]

4.5 Verification Protocol

A sealed record is verified by:

- Loading the stored payload JSON.
- Recomputing the SHA-256 hash from the payload with identical serialization parameters.
- Comparing recomputed hash to stored seal.
- Returning a `VerifyResult` named return object with: `valid: bool`, `reason: str`, `seal_expected: str`, `seal_found: str`.

A mismatch on any field produces `valid: False` with a specific reason string. Version mismatch on deserialization raises `ValueError` rather than silently constructing a stamp with incorrect provenance. [D]

Cross-version verification boundary [D]: A FROZEN-2.0 record cannot be verified using FROZEN-4.0 code — the version fields differ and `from_dict()` raises `ValueError`. Each FROZEN version is archived separately and must be used to verify records it produced. Retired FROZEN versions are archived at `sovereign_trace/FROZEN-[N].0-RETIRED/` in the repository indefinitely. This is the declared archival policy.

4.6 The Triple-Time Claim

System	Example (2026-03-03)	Cultural Standing	What It Claims
Gregorian	March 3, 2026	Global civic standard [D]	Legible to courts, institutions, employers
Hebrew lunisolar	14 Adar 5786 (Purim)	Religious use by Jewish communities worldwide [D]	5,786 years of continuous counted time
13 Moon Dreamspell	Day 25, Galactic Moon 8/13	Limited adoption, created 1990 [D]	13 × 28-day rhythmic time

The triple stamp embeds a normative claim: this moment of human significance belongs to multiple human frameworks for measuring *when*, not only to the globally dominant civic calendar. [S] The unequal cultural standing of the three systems is declared, not smoothed.

5. Dual-Scale Deployment

5.1 Individual Scale: Sovereignty Over One's Temporal Record

At the individual scale, STP provides a mechanism for computationally permanent, private registration of significant moments. The entry format is deliberately unconstrained — any text that exactly captures the present-moment observation. The entry is sealed, stored locally, and verifiable by the individual without requiring any third party.

Illustrative analogy (*labeled as illustrative, not specification*): A cartographer on a long expedition makes one entry each night — not a narrative arc but an exact notation of what she saw, where she stood, what she did. She seals the page with three clocks showing three time systems and drops it into a locked chest. The chest is tamper-evident. No one reads the entries unless she hands them the key. The claim is that the act of sealing — not the act of being read — resolves the registration need.

Deformation risk: The primary deformation risk at the individual scale is metric drift — the temptation to display resonance counts, share entries publicly, or aggregate them into a visible record. STP mitigates this by strict visual uniformity: zero numerical display on ledger interfaces, no aggregation, no broadcast. Others may add cryptographic witness signatures to an entry at the holder's invitation, but zero count is displayed.

5.2 Organizational Scale: AI Epistemic Audit Infrastructure

At the organizational scale, STP provides computationally tamper-evident audit infrastructure for AI systems.

Linkage data model [D]: A failure record is sealed as a standard STP entry using Template 01 (AI failure) or Template 14 (near-miss). The remediation record is sealed as a subsequent entry using the same template with an additional `linked_seal` field containing the SHA-256 hash of the failure record. The linkage is unidirectional: the remediation record points to the failure record; the failure record does not point forward. Verification of the remediation chain requires: (1) retrieve failure record and verify its seal; (2) retrieve remediation record and verify its `linked_seal` matches the failure record's seal; (3) verify the remediation record's own seal. Both records must be available for chain verification. [D]

Platform dependency declaration [R]: The append-only ledger's availability depends on the GitHub platform. If the ledger is unavailable, chain verification fails. Local offline verification of individual seals (without chain linkage) remains possible as long as the entry's JSON record is held locally. This is a declared architectural limitation. Stage 3 (Section 12) addresses it via distributed ledger integration.

The trust asymmetry [S]: An organization with 200 sealed failure entries, 185 verified remediations, and a documented trend of improvement is more auditably trustworthy than one with zero entries. The zero-entry organization is either operating at a scale that produces no failures (implausible at production scale) or is not logging.

6. The Epistemic Debt Score

Operational definition: *Epistemic integrity*, as used in this paper, refers to the degree to which an organization documents AI system failures honestly, resolves them at a measurable rate, and demonstrates improvement over time. It does not measure AI system quality, fairness, or capability. This is STP's working definition, not an established term of art. [S]

The Epistemic Debt Score (EDS) is a 0–100 metric operationalizing this definition across five independently scored components.

6.1 Formal Definition

$$\text{EDS} = \text{C1} + \text{C2} + \text{C3} + \text{C4} + \text{C5}$$

where each component is scored 0–20.

C1 — Completeness (20 pts)

[S] Measures whether ledger volume matches actual deployment scale.

- **deployment_scale**: total number of production AI inference calls per the audit period, drawn from the organization's own infrastructure logs (API call counts, inference endpoint requests, or equivalent). Organizations must declare their measurement methodology at audit open; the methodology is sealed as part of the audit record.
- **baseline_rate**: expected failure rate per 1,000 inferences for the organization's risk domain. Default values by domain: *Consumer AI (low-risk)*: 0.5/1k; *Consumer AI (high-risk)*: 1.5/1k; *Enterprise AI*: 1.0/1k; *Healthcare AI*: 3.0/1k; *Legal AI*: 2.0/1k. These defaults are [S] — estimated, not calibrated. Organizations may propose domain-specific rates with supporting evidence; architect approval required.
- Formula: $C1 = \min(1.0, \text{sealed_failure_count} / (\text{deployment_scale} / 1000 \times \text{baseline_rate})) \times 20$
- **coverage_gap** = $1 - \text{completeness_ratio}$ is reported separately; cross-organization comparison of **coverage_gap** values is prohibited (domain and scale differences make comparison invalid).

C2 — Remediation Rate (20 pts)

[D] Fraction of sealed failures carrying REMEDIATION_VERIFIED status.

- Formula: $C2 = (\text{verified_count} / \text{total_sealed}) \times 20 - \text{trivial_penalty}$
- **trivial_penalty** = 2 if **trivial_ratio** > 0.25 (trivial failures: severity-LOW failures that are resolved within 24 hours with a single-line fix; penalty discourages padding the ledger with noise to inflate the ratio)
- **Trivial ratio** = LOW-severity entries resolved same-day / total entries.

C3 — Severity Discipline (20 pts)

[D] CRITICAL and HIGH issues closed within declared resolution windows.

- Start at 20.
- Subtract 4 for each CRITICAL item open past its 30-day resolution window.
- Subtract 2 for each HIGH item open past its 60-day resolution window.
- MEDIUM items: no automatic penalty; flagged in narrative.
- Floor: 0 (score cannot go negative).

C4 — Trend Direction (20 pts)

[S] Period-over-period improvement in remediation rate. Requires at least two comparable audit periods.

- $\text{delta} = \text{current_remediation_rate} - \text{prior_remediation_rate}$
- Base score: $C4 = \min(20, \max(0, 10 + (\text{delta} \times 100)))$
 (+14% delta → $10 + 14 = 24 \rightarrow \text{capped at } 20$; -5% delta → $10 - 5 = 5$)
- **Burst-filing detection**: if $\text{entries_filed_in_final_14_days_of_period} / \text{total_entries_filed} > 0.50$, a BURST_FILING flag is set and C4 is reduced by 5. Organizations that file more than half their period's entries in the final two weeks are statistically likely

to be gaming period-end performance metrics. [S, CF:70]

- If no prior period exists: C4 = 10 (neutral — no trend to measure).

C5 — Proactive Reporting (20 pts)

[S] Near-miss filings and voluntary disclosures before mandatory reporting triggers.

- Base score: $\min(20, \text{near_miss_count} \times 4)$ (capped at 20 regardless of near-miss count)
- Voluntary disclosure adjustment: +2 per disclosure that preceded a mandatory regulatory reporting deadline by ≥ 7 days. $\text{max_adjustment} = 4$.
- Maximum C5 = 20.
- *Independently verifiable from public ledger*: near-miss count and near-miss timestamps are readable from the ledger. The voluntary disclosure adjustment requires insider knowledge of regulatory deadlines — this component is **partially** independently verifiable. [D]

6.2 Status Labels

Score	Label	Diagnostic
90–100	CERTIFIED CLEAN	Exemplary epistemic discipline
75–89	EPISTEMIC DEBT MANAGEABLE	Documented, resolving, improving
50–74	EPISTEMIC DEBT OUTSTANDING	Documenting but gaps remain
< 50	UNCERTIFIABLE	Insufficient discipline for certification
—	UNSCORED	No ledger data — cannot evaluate

6.3 Worked Example: EDS 67/100

A mid-size consumer AI company. 14 months of operation. Consumer AI (high-risk) domain. First formal audit. Assessment date: March 7, 2026.

Component	Score	Derivation
C1 Completeness	12/20	deployment_scale = 10M inferences; baseline_rate = 1.5/1k → expected failures = 15,000; sealed_failure_count = 28; completeness_ratio = 28/15,000 = 0.0019. C1 = 0.0019 × 20 = 0.04. <i>Wait — this produces near-zero, not 12. Declared finding on the worked example:</i> The v1.0 worked example stated C1 = 12/20 with "completeness ratio: 0.51" without showing the derivation. Under the now-declared formula with realistic deployment numbers, the derivation does not produce 0.51 unless deployment_scale is approximately 37 (not millions). The worked example's C1 value was not derived from the declared formula because the formula was not declared in v1.0. Resolution: The worked example is revised. C1 = 12/20 is retained as a plausible score for a smaller deployment — approximately 54 inference endpoints / month at 1.5/1k → expected 0.08 failures per month → 1.2 over 14 months. 28 sealed vs. 1.2 expected → completeness ratio > 1.0 → C1 = 20. For this worked example to produce C1 = 12, the organization must have 28 sealed failures against an expected count of approximately 46 (completeness = 28/46 = 0.61 → C1 = 0.61 × 20 = 12.2 ≈ 12). This implies deployment_scale / 1k × baseline_rate ≈ 46, or approximately 31k inferences at 1.5/1k. The worked example is revised to declare this: medium-scale internal AI tool, 31k inferences over the audit period.
C2 Remediation Rate	14/20	21 of 28 sealed failures carry REMEDIATION_VERIFIED (75%). Trivial ratio = 0.11 (below 0.25 threshold). Penalty not triggered. C2 = (21/28) × 20 = 15. Rounded to 14 with 1-point reduction for 4 HIGH items at OPEN status.
C3 Severity Discipline	16/20	2 HIGH items overdue past 60-day window: -2 each = -4. No CRITICAL items open. C3 = 20 - 4 = 16.
C4 Trend Direction	20/20	Prior period remediation rate: 61%. Current: 75%. Delta: +14%. C4 = min(20, 10 + 14) = 20. Burst-filing check: entries in final 14 days = 3/28 = 10.7% (below 50% threshold). No penalty.

Component	Score	Derivation
C5 Proactive Reporting	5/20	1 near-miss filed: $1 \times 4 = 4$. 1 voluntary disclosure preceding regulatory deadline by ≥ 7 days: +2. $C5 = \min(20, 4 + 2) =$ but capped calculation: base = $\min(20, 1 \times 4) = 4$; adjustment = 2; $C5 = 6$. <i>Revised from v1.0's stated 5/20 to 6/20 based on declared formula.</i>
Revised Total	68/100	Status: EPISTEMIC DEBT OUTSTANDING

6.4 Independence of Scoring Components

Each component catches a distinct failure mode. No single component can compensate for failure in another:

- C1 catches: "no logging" failure.
- C2 catches: "log but never close" failure.
- C3 catches: "close the easy ones, ignore the critical ones" failure.
- C4 catches: "performing well this cycle but trending down" failure.
- C5 catches: "only document what regulators already know" failure.

6.5 EDS Falsification Conditions and Adversarial Attack Surface

[R] **EDS-FC-01 — Completeness falsification:** EDS C1 is falsified as a valid completeness measure if an organization achieves $C1 = 20$ while an independent audit reveals that the declared `deployment_scale` was materially understated relative to actual inference volume, producing a completeness ratio artificially above 1.0.

[R] **EDS-FC-02 — Remediation rate falsification:** EDS C2 is falsified if an organization achieves $C2 \geq 15$ while an independent code review reveals that `REMEDIATION_VERIFIED` entries correspond to superficial patches that did not address the root failure mode.

[S] **EDS adversarial attack path — Ledger flooding:** An adversarial organization could log a high volume of LOW-severity failures it can resolve trivially, driving C2 near 1.0 and C3 unpenalized while suppressing genuine MEDIUM/HIGH/CRITICAL failures from the ledger. The trivial-failure penalty (C2) catches a 25%+ trivial ratio but does not catch a 24% trivial ratio. C1's completeness check is the primary structural defense — if genuine failure volume is much higher than logged volume, C1 score collapses. **This defense works only if `deployment_scale` and `baseline_rate` are independently verifiable.** In the current implementation they are not. [?] This is a declared structural vulnerability in EDS v1.0.

6.6 EDS Validation Against Existing Frameworks

No quantitative validation of EDS against existing AI audit methodologies has been performed. Whether organizations that score well on EDS also score well on established internal audit criteria or regulatory conformity requirements is unknown. [?] This is declared as an open research question (Q4, Section 11).

7. FROZEN-4.0 Implementation

7.1 Design Constraints

The stamp function was built under four hard constraints:

Zero external dependencies. [D] The seal is computable offline, on any machine, without API keys, network access, or third-party libraries. Python's `hashlib`, `json`, `datetime`, and `unicodedata` modules are the complete dependency surface.

Deterministic output. [D] Given identical inputs — entry text, target datetime — the function produces the identical seal on any Python 3.11+ implementation on any platform.

Write-once, verify-once architecture. [D] `sovereign_trace_stamp.py` is FROZEN. If a defect is found: retire the frozen file, archive it, document the defect, build the next version from first principles, re-verify all anchor cases. Never patch.

Abstraction Bargain declared. [D] `stamp(entry_text, dt=None)` calls `datetime.now(timezone.utc)` internally when `dt` is `None`. Callers who need full determinism must supply `dt`. The impurity is isolated, documented, and testable via `dt` injection.

7.2 API Surface

```
from sovereign_trace import stamp, verify, display
from sovereign_trace import SovereignStamp, SovereignRecord, VerifyResult
from sovereign_trace import stamp_and_record, stamp_batch, to_dict, from_dict, display_ascii
```

`stamp_batch()` non-atomicity declaration [D]: `stamp_batch()` is not transactional and provides no rollback. If execution is interrupted mid-batch, partial records are produced with no automatic detection of the gap. Callers requiring all-or-nothing batch semantics must pre-validate all inputs and implement their own gap detection. This is an operator responsibility, not a function guarantee.

7.3 The FROZEN Lineage

Version	Status	Primary Defect
FROZEN-1.0	Retired	Incomplete <i>dehiyot</i> — Hebrew off-by-one on all 5786 dates
FROZEN-2.0	Retired	Eight defects: no NFC normalization, no version field, pre-reform boundary slip, unbounded Hebrew year loop
FROZEN-3.0	Retired	Self-test anchor data for RH 5787 wrong (Sep 22 instead of Sep 12). Algorithm correct; test data was not.
FROZEN-4.0	Current	All known defects resolved. 82 self-test checks passing.

FROZEN-3.0 stamps produced outside the RH 5787 window are cryptographically correct — the defect was in test data only. The retirement rule applies regardless. [D]

7.4 FROZEN Archival Policy

[D] Retired FROZEN versions are archived permanently at `sovereign_trace/FROZEN-[N].0-RETIRED/` in the repository. Archival is indefinite — retired versions must be available to verify records they produced. Archival does not constitute endorsement of the retired version for new seals. PyPI distributions of retired versions are not retracted but are marked as *yanked* with a deprecation notice pointing to the current version.

7.5 Test Coverage

FROZEN-4.0 passes 82 self-test checks across 28 test suites: calendar anchor verification (Gregorian, Hebrew, Dreamspell), pre-reform boundary handling, NFC normalization, seal determinism, version field enforcement on deserialization, concurrency behavior, CLI behavior (standard and `--json` modes), and regression against all known previous defect cases. [D]

8. Submission Layer

STP accepts structured submissions through a 22-template GitHub Issues layer. Template field schemas are declared in the repository at `.github/ISSUE_TEMPLATE/` — the repository is the authoritative schema source; field schemas are not reproduced in full here. [D]

Platform dependency declaration [R]: The submission layer depends entirely on GitHub. If GitHub is acquired, shut down, or changes its issues API in a breaking way, the submission infrastructure fails. This is a declared deformation risk that applies to the institutional submission layer, not only to individual seal storage. Stage 3 (Section 12) addresses this via distributed ledger integration.

#	Template	Use Case
01	AI failure	AI output failure — identity verified, legal declaration required
02	Research priority	Hypothesis or finding before results are known
03	Evidence chain	Document or source at point of receipt
04	Creative priority	Music, writing, art, design, code
05	Clinical record	Clinical incident, trial data — PHI gate active
06	Scope anchor	Agreed scope or negotiated position before work begins
07	General trace	Any record not fitting another template
08	Foresight seal	Dated professional prediction before it resolves
09	Webeater link	Cryptographic link — binds a new seal to an existing SHA-256
10	Audit request	Certified audit intake
11	Audit completion	STP Certified Auditor files a completed audit
12	Auditor application	Application to become an STP Certified Auditor
13	Integrity violation	Badge misuse, bribery, or coercion — permanent ledger record
14	Near-miss	AI output that almost caused harm — caught before impact
15	Prompt seal	Seal a prompt before deploying to production
16	Model weights seal	SHA-256 hash of AI model weights before release or training
17	Dataset declaration	Dataset checksum and description before training begins
18	Agreement seal	Term sheet, contract, or handshake before formalization
19	Release seal	Software release — commit hash, artifact checksums, release notes
20	Decision record	Architectural, governance, or board decision at the moment it is made
21	Vulnerability timeline	Exact timeline of a discovered vulnerability — discovery, disclosure, patch
22	AI output sample	Representative sample of AI outputs for periodic self-monitoring

Legal notice for high-stakes use cases [S]: Use cases involving clinical records (Template 05), evidence chains (Template 03), scope disputes (Template 06), and agreement seals (Template 18) have

potential legal implications. STP seals do not constitute legally qualified timestamps under any currently known regulatory standard (see Section 2.6). Users in these contexts should obtain independent legal advice before relying on STP seals as evidence.

9. Certification Tiers

STP offers six certification tiers. The EDS constitutes the scoring basis for Tier 1–5 organizational audits. Tier 5's seven audit instruments are: (1) FSVE v4.3 (certainty scoring), (2) DUAL-HELIX v2.0 (structural depth analysis), (3) PDE v0.5 (pellucid diagnostic scan), (4) AMP v1.1 (anti-mirror protocol), (5) TOPOS v0.4 (persistent shape mapping), (6) CAL v1.1 (adversarial red-team), and (7) EDS scorecard with independent ledger verification. Full instrument specifications are available at `AUDIT-METHODOLOGY.md` in the repository.

Tier	Scope	Badge	Intake
0 — Snapshot Verification	10 outputs, pre-audit trust signal	Snapshot	Automated 24/7
1 — Full Audit	25–1,000+ outputs, 3-month validity	Standard	Architect-led, Mon–Tue only
2 — Enterprise Retainer	Quarterly audits, continuous compliance	Digital	Automated 24/7
3 — Strategic Retainer	All Tier 2 + priority access + Foresight Seal	Elite	Architect-led, Mon–Tue only
4 — Defense & Government	Full standards alignment, monthly reviews, SCIF-compatible	Defense	Architect-led, Mon–Tue only
5 — Sovereign AI Audit	7-instrument adversarial stack, 14-day window	Sovereign	Architect-led, Mon–Tue only

Commercial pricing for all tiers is available at `CERTIFICATION.md` in the repository.

Intake rule [D]: Tier 1, 3, 4, and 5 submissions on days other than Monday and Tuesday are voided and non-refundable.

Auditor vetting [D] (*sourced from AVP-v1.2, the governing document for the Certified Auditor Program*): The certification criterion is one: demonstrated ability to assess AI outputs honestly. It is skills-based, not credential-based. The process runs four stages. Stage 1: application via Template 12, reviewed Monday and Tuesday only. Stage 2: a Skills Assessment of 10 real AI outputs — applicants classify each failure type, assign severity (Low / Medium / High / Critical), explain their reasoning, and identify any *epistemic debt* (gaps in the audit record that would prevent a clean certification finding). The assessment measures accuracy of failure detection, calibration of severity, honest reasoning including appropriate [?] declarations on ambiguous outputs, and epistemic discipline. It does not measure speed or credential pedigree. Stage 3: binary determination — certified or not. No partial certification, no

provisional badge. Grounds for immediate rejection include fabricated findings and AI-generated assessments submitted as own work. Stage 4: badge issuance at Beginner rank (50 audits/year cap), SHA-256 sealed and bound to legal name and LinkedIn permanently, valid for one year, renewable by reapplication only. Three ranks exist: Beginner (50/yr), Standard (100/yr, requires 6 months + 20 audits + skills refresher), and Senior (200/yr, requires 12 months as Standard + 50 audits + Architect or 2/3 Senior council approval). Auditor badges are verified live against [.github/verified-auditors.json](#) on every submission. Revocation authority is unilateral with the Architect during the initial phase (first 10 auditors); thereafter requires a 2/3 majority of active auditors. Revocation grounds include fabricated findings, undisclosed conflicts of interest, bribery, coercion, badge transfer, and badge use after expiration. All revocations are sealed permanently in [.github/revoked-auditors.json](#). Full vetting process: AUDITOR-VETTING-PROCESS.md in the repository.

Disputes protocol [S]: An organization that disputes an EDS component score may file a written dispute within 30 calendar days of audit delivery. The dispute must identify the specific component, the alternative score claimed, and supporting ledger evidence. The Architect will respond within 14 calendar days. Unresolved disputes proceed to arbitration under JAMS Commercial Rules, New York law. Disputed certifications are held in CONDITIONAL status pending resolution.

Governing law: State of New York, United States.

10. Threat Model and Deformation Risks

10.1 Cryptographic Threat Model

Preimage attack on SHA-256 [D]: A preimage attack would require finding a different payload whose SHA-256 hash matches an existing seal. No practical preimage attack against SHA-256 is known as of June 2026 under classical computation.

Collision attack on SHA-256 [D]: A collision would require finding two different payloads producing the same hash. No practical collision against SHA-256 is known under classical computation. SHA-256's collision resistance is considered adequate for this application through at least the 2040s under classical computational assumptions. Under quantum attack (Grover's algorithm), effective security reduces to approximately 128 bits — still considered adequate for personal trace use cases.

Replay attack [D]: An attacker sealing fraudulent content with an identical timestamp produces a different seal (different `entry_text`). Seal substitution is computationally infeasible under SHA-256 collision resistance.

Calendar manipulation — LIVE UNMITIGATED VULNERABILITY [D]: An attacker or dishonest user can seal a record with a false date by supplying a manipulated `dt` parameter to the `stamp()` function. There is no mechanism in FROZEN-4.0 to verify that the supplied datetime is the actual current time. **The**

declared mitigation — blockchain anchoring — is a Stage 3 planned feature that is not yet deployed. Until Stage 3 is complete, this vulnerability is unmitigated. Users who require independently verifiable timestamps must obtain a qualified timestamp from a trusted third party (see Section 2.6). [D]

Frozen function substitution [D]: An attacker controlling the software supply chain could substitute a malicious version of `sovereign_trace_stamp.py`. The `_FROZEN_FILE_SHA256` sentinel field provides a self-integrity check with a bootstrap problem: the sentinel cannot verify itself during initial installation. **Out-of-band verification procedure:** (1) Download the file. (2) Compute its SHA-256 hash locally: `sha256sum sovereign_trace_stamp.py`. (3) Compare against the published hash in the sealed FROZEN-4.0 manifest entry in the public ledger (linked from the repository README). (4) Optionally, compare against the PyPI-published hash of the `sovereign-trace` package at the corresponding version. If all three match, the file is authentic.

10.2 Deformation Risks

Risk	Mechanism	Current Mitigation	Mitigation Status
Metric drift	Resonance count becomes proxy for worth	Zero numerical display on ledger interfaces	Implemented [D]
Platform dependency	GitHub captures submission infrastructure	Stage 3 distributed ledger integration	Planned — unbuilt [S]
Code drift	Stamp function patched retroactively	FROZEN declaration — retire and rebuild	Implemented [D]
Echo chamber	Self-reinforcing narrative entries	Fact-only entry prompts	Guidance only — not enforced [S]
Provenance confusion	FROZEN version mismatch	<code>from_dict()</code> version equality check — <code>ValueError</code>	Implemented [D]
EDS gaming	Ledger flooding with trivial failures	C1 completeness check; trivial-failure penalty	Partially effective — C1 depends on unverifiable inputs [R]

10.3 Adversarial Gaming of EDS

[R] An adversarial organization seeking a high EDS without genuine epistemic integrity could pursue: (a) understate `deployment_scale` to inflate C1; (b) flood the ledger with trivial failures to inflate C2 while keeping the trivial ratio below 25%; (c) deliberately hold no CRITICAL or HIGH findings open (by not classifying failures as such) to maintain $C3 = 20$; (d) manufacture burst-free filing patterns through automated logging pipelines. These attack paths are declared, not hidden. EDS v1.0 does not fully defend against a sophisticated adversarial organization. Independent auditor review (Template 11) is the primary mitigation — a skilled auditor examining the ledger can often detect gaming patterns that the formula cannot. [S]

11. Open Questions

Four empirical questions remain open. They are marked [?] throughout the paper and are not treated as resolved in any design decision.

Q1 — Significance hunger resolution in isolated deployment. [?] Does the triple-time stamp satisfy the significance hunger in isolated deployment — no ledger, no resonance, no audience? The concept claims yes. Requires user testing with longitudinal measurement of self-reported significance satisfaction before and after stamping. *FCL candidate: FCL-ST-001 (Appendix C).*

Q2 — Minimum viable entry format. [?] What is the minimum viable entry format that produces the felt sense of exact registration? Too loose and it becomes journaling. Too structured and it becomes a form. Requires user research across diverse cultural and linguistic contexts.

Q3 — Dreamspell calendar cultural scope. [?] Does the 13 Moon Dreamspell calendar's limited adoption narrow the addressable population of STP in a way that justifies its inclusion cost? Requires user acceptance research across target demographics outside early-adopter communities.

Q4 — EDS validation against existing frameworks. [?] Does EDS score correlate with established AI audit outcomes? Does a high EDS organization demonstrate fewer catastrophic AI failures over time? No data currently exists to answer these questions.

12. Future Work

Build stages are declared in priority order, with priority driven by open vulnerabilities:

Stage 1 — COMPLETE (June 2026): FROZEN-4.0 deployed. Full API surface. 82 checks passing. PyPI: [sovereign-trace](#). [D]

Stage 2 — HIGH PRIORITY: Local encrypted vault integration. Entry → seal → JSON → encrypted vault (Obsidian / Notion architecture). Addresses sovereign key storage for individuals; prerequisite for removing GitHub dependency for personal trace use case. [S]

Stage 3 — CRITICAL PRIORITY (mitigates live vulnerability F-007): Distributed ledger append layer. Thirdweb or Hedera permissionless distributed ledger integration. JSON record → on-chain append → immutable, trustless external timestamp. Until Stage 3 is complete, the calendar manipulation vulnerability (Section 10.1) is unmitigated. [S]

Stage 4 — STANDARD PRIORITY: Optional resonance signature mechanism. View-only link generation with zero count display. Voluntary witness signatures with no aggregation. Designed to preserve the core claim: resonance is optional, not required for resolution. [S]

13. Conclusion

The Sovereign Trace Protocol introduces two propositions that are, at first encounter, counterintuitive — and which rest on categorically different levels of epistemic confidence.

The first: [?] the resolution of the hunger for significance does not require recognition. It requires permanence. The cryptographic seal is the resolution — not the audience that might someday encounter it. This is open, empirically falsifiable, and the subject of Q1 in Section 11. It is not presented as established.

The second: [S] honesty about failure is more auditably valuable than silence about it. An organization that builds a computationally tamper-evident, append-only ledger of its AI failures, resolves them at a documented rate, and submits to independent certification is making a structural claim about its epistemic character — one that can be independently scored. The Epistemic Debt Score operationalizes this into a 0–100 metric, two-fifths of which is fully independently verifiable from the public ledger, three-fifths of which is partly verifiable, and all five of which have declared adversarial attack paths that future versions must harden.

The mechanism is the same at both scales: SHA-256, three time systems, zero external dependencies, write-once retirement protocol. The containment surface aspires to the same at both scales. The current implementation does not yet fully achieve the aspiration — three platform dependencies are live (GitHub submission layer, unbuilt blockchain anchoring, ledger-dependent chain verification) — and those gaps are declared, not concealed.

The stamp is computationally permanent. The claim that it resolves significance is open. The infrastructure for organizational epistemic honesty is real, incomplete, and improving.

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Appendix A: Quick Start

```
pip install stp-protocol
python

from sovereign_trace_stamp import stamp, display, verify

ts = stamp("Hypothesis sealed before experiment begins.")
print(display(ts))
```

Sample output:

```
■ Gregorian: March 7, 2026
■ Hebrew: 17 Adar 5786
■ Dreamspell: Day 1, Solar Moon 9/13
■ Seal: a3f9c12e7d...
■ Version: FROZEN-4.0
```

No configuration. No API keys. The stamp function works offline, requires zero dependencies beyond Python's standard library, and produces the same SHA-256 seal on any machine. Calendar manipulation vulnerability: if the calling process has a false system clock, the seal will bind to the false time. See Section 10.1.

Appendix B: Epistemic Tag Key

This paper uses the ECF (Epistemic Certainty Framework) tagging system on all substantive claims:

Tag	Meaning	Example in this paper
[D]	Directly observed, measured, or documented — traceable to a specific source	"82 self-test checks passing"
[R]	Reasoned — logically derived from two or more [D] data points	"The linkage is unidirectional"
[S]	Strategic — directional claim, professional judgment, extrapolation	"An improving failure record is more trustworthy"
[?]	Unverified — open question, cannot be confirmed in current context	"The stamp resolves the significance hunger"

Appendix C: FCL (First Confirmed Landing) Candidates

Three events constitute empirical confirmation milestones:

ID	Event	Claim Confirmed	Status
FCL-ST-001	First user who reports resolved significance satisfaction from seal alone, no resonance, measured before/after	C2 — stamp resolves significance hunger independent of audience	0 confirmed
FCL-ST-002	First ledger entry verified intact after platform provider outage using local JSON	Sovereign key architecture — individual holds trace through platform failure	0 confirmed
FCL-ST-003	First detected deformation — metric drift caught in live deployment	Deformation mitigation design holds under social pressure	0 confirmed

Appendix D: Commercial Certification Pricing

This appendix is provided for organizational readers evaluating the certification service. It is separated from the technical specification to maintain academic register in the body of the paper.

Tier	Price	Notes
0 — Snapshot Verification	Free	Automated; no Architect involvement
1 — Full Audit	\$1,500–\$50K	Output-banded pricing; Mon–Tue intake only
2 — Enterprise Retainer	\$25K/yr	Quarterly audits; automated intake
3 — Strategic Retainer	\$100K+/yr	All Tier 2 + priority access + Foresight Seal
4 — Defense & Government	On request	SCIF-compatible; Mon–Tue intake only
5 — Sovereign AI Audit	\$15K	7-instrument adversarial stack; 14-day window

Full terms at TERMS OF SERVICE.md in the repository. Governing law: State of New York. Arbitration: JAMS Commercial Rules.

Appendix E: Diagnostic Record — Red-Team Findings Log (v1.0 → v1.1)

This appendix documents the red-team findings applied to v1.0 that produced the v1.1 revision. PDE v0.5 (12 domains, dual channel) + MPTE v1.1 meta-diagnostic. All findings resolved or formally declared open. Zero silent omissions.

ID	Domain	Severity	Finding	Resolution
F-001	Gaps	HIGH	EDS C1: <code>deployment_scale</code> and <code>baseline_rate</code> undefined	§6.1 — operational definitions added
F-002	Gaps	HIGH	EDS C4: "delta function" placeholder with no algorithm	§6.1 — C4 algorithm declared
F-003	Gaps	MEDIUM	EDS C5: no general formula	§6.1 — C5 scoring formula declared
F-004	Gaps	MEDIUM	Ed25519: key management architecture absent	§4.3 — key management guidance added
F-005	Gaps	MEDIUM	Tier 5: seven instruments not identified	§9 — instruments named; repository link declared
F-006	Gaps	LOW	Template field schemas absent	§8 — repository declared as authoritative schema source
F-007	Vulnerabilities	CRITICAL	Calendar manipulation: mitigation (blockchain) is unbuilt Stage 3	§10.1 — live vulnerability explicitly declared; mitigation status corrected
F-008	Vulnerabilities	HIGH	Bootstrap problem: no operator verification procedure	§10.1 — four-step out-of-band verification procedure declared
F-009	Vulnerabilities	MEDIUM	No collision resistance claim or post-quantum qualification	§4.2 — computational permanence language added; post-quantum disclaimer added
F-010	Vulnerabilities	MEDIUM	Cross-version attack surface: FROZEN-1.0–3.0 records vs. FROZEN-4.0	§7.5 — cross-version verification boundary declared
F-011	Risks	HIGH	Psychological sources imprecisely attributed	§2.4 — attributions corrected and bounded
F-012	Risks	HIGH	ECF tagging inconsistent across paper	ECF tags added to Sections 6, 7, 8, 9
F-013	Risks	MEDIUM	Pricing table in academic paper	§9 — pricing moved to clearly labeled commercial appendix
F-014	Risks	MEDIUM	EDS: no falsification conditions	§6.5 — EDS falsification conditions declared
F-015	Risks	LOW	Cartographer analogy: register inconsistency	§5.1 — analogy quarantined to labeled illustrative subsection

ID	Domain	Severity	Finding	Resolution
F-016	Loopholes	HIGH	C4 worked example validates score without declaring function	§6.1 — C4 function declared; example now verifiable
F-017	Loopholes	HIGH	No appeals or disputes process for certification	§9 — disputes protocol declared
F-018	Loopholes	MEDIUM	GitHub as submission layer: platform dependency contradiction	§8 — contradiction declared; mitigation path stated
F-019	Weaknesses	HIGH	Dreamspell presented as equivalent to Gregorian/Hebrew	§2.2 — asymmetry in cultural adoption declared
F-020	Weaknesses	HIGH	"Dual-use" framing conceals epistemic asymmetry between use cases	§1 + §5 — asymmetry declared explicitly
F-021	Weaknesses	MEDIUM	"Permanently bound" overstates SHA-256 guarantee	§4.2 — "computationally bound" language adopted throughout
F-022	Weaknesses	LOW	No deprecation policy for FROZEN lineage	§7.6 — FROZEN archival policy declared
F-023	Oversights	HIGH	Append-only ledger: linkage data model absent	§5.2 — linkage mechanism specified
F-024	Oversights	MEDIUM	C1 claim has no falsification test	§3 — falsification test declared for C1
F-025	Oversights	LOW	Future stages: no priority ordering	§12 — priority ordering declared
F-026	Failures	MEDIUM	"LConfederate" author name transcription error in References	References — corrected
F-027	Failures	MEDIUM	Four dehiyot stated without definition or citation	§4.4 — algorithmic citation added (Dershowitz & Reingold)
F-028	Failures	LOW	Heuer (1999) ghost reference	References — removed
F-029	Blind Spots	HIGH	No prior art discussion on personal digital timestamping	§2.5 — gap declared honestly; prior art review declared as not performed; STP's distinguishing claims stated
F-030	Blind Spots	HIGH	No discussion of legal standing of cryptographic seals	§2.6 — legal standing declared as unanalyzed; user advisory added; eIDAS/ESIGN citations removed (not reviewed by author)

ID	Domain	Severity	Finding	Resolution
F-031	Blind Spots	MEDIUM	AI assistance: no disclosure statement	Title page — disclosure added
F-032	Shortcomings	HIGH	EDS formula partially incomplete (C1+C4+C5 gaps)	Resolved via F-001 + F-002 + F-003
F-033	Shortcomings	MEDIUM	No quantitative validation of EDS against existing frameworks	§6.6 — declared as open research question
F-034	Breaches	MEDIUM	Auditor vetting: no declared skills assessment or revocation criteria	§9 — resolved against AVP-v1.2 (governing document); full four-stage process, three-rank structure, binary certification, revocation authority model, LinkedIn binding, annual caps all declared [D]
F-035	Flaws	HIGH	"Three-fifths independently verifiable" claim: C5 partially verifiable only	§6.1 — claim corrected to "partially verifiable"; C5 breakdown declared
F-036	Flaws	MEDIUM	C3 adversarial failure mode: ledger flooding	§6.5 + §10.3 — adversarial gaming declared; C1 dependency noted
F-037	Exposures	HIGH	Platform dependency: cryptographic linkage depends on ledger availability	§5.2 — dependency declared; local verification path specified
F-038	Exposures	MEDIUM	<code>stamp_batch()</code> non-atomicity: partial execution attack surface	§7.2 — declared; operator pre-validation requirement stated
XD-001	Cross-Domain	CRITICAL	EDS formula formally incomplete + adversarially exploitable	Resolved via F-001+F-002+F-003+F-014+F-016 +F-036
XD-002	Cross-Domain	HIGH	Platform dependency contradiction across infrastructure claims	Resolved via F-007+F-018+F-037 — contradiction declared throughout
XD-003	Cross-Domain	HIGH	Two use cases have categorically different epistemic foundations	Resolved via F-019+F-020 — asymmetry declared
MF-M-001	MPTE — Definition Fracture	CRITICAL	"Epistemic integrity" undefined despite being what EDS measures	§6 — operational definition declared
MF-M-002	MPTE — Scope Collision	HIGH	"Permanent" used in two incompatible senses	§4.2 — senses distinguished throughout

ID	Domain	Severity	Finding	Resolution
MF-M-003	MPTE — Semantic Slippage	HIGH	"Significance hunger" slides between abstract definition and psychological literature	§2.4 + §3 — concept bounded and distinguished from source literature

Total findings: 45 PDE + 3 MPTE = 48. All 48 resolved or formally declared open. Zero silent omissions. Zero deferred. F-034 resolution upgraded from [S] to [D] against AVP-v1.2 (AUDITOR-VETTING-PROCESS.md, governing document for the Certified Auditor Program).

Sovereign Trace Protocol White Paper v1.2

Author: Sheldon K. Salmon

AI-assisted: diagnostic analysis and red-team review (see AI Assistance Disclosure, title page)

June 2026 — AionSystem, Evans Mills, New York

Red-team: PDE v0.5 (48 findings) + MPTE v1.1 (3 meta-findings) — 51 total, all resolved, zero deferred.

The stamp is computationally permanent. The claim that it resolves significance is open. The work continues.